Homework 4

1. Review code pointer1.c and write your answer to the following questions in file 4_1.c:

```
int a=5, b, *p=&a;
```

Assuming that the lvalue of a, b, and p are: `fb7c`, `fb78`, `fb74`, write down the rvalue and lvalue of a, b, and p after each statement (answer to the first is shown as an example):

```
p = &a;        // a: (5, fb7c); b: (unknown, fb78); p: (fb7c, fb74)
b = a * *p;
*p = b + a;
b = (*p)++;
a = *p++;
p = &b;
a = *p;
b = a++;
p = p + 32;
```

3. Complete 4.3.c in public/directory so it will re-produce exactly what 4.3.out does. Run 4.3.detail.out with the same seed value for more details.
4. Write a complete code to estimate the area enclosed by curve $y=x^2$, curve $y=x^3$, and vertical line x=a, where integer a > 1. The value of a and the number of sample points will be provided from the command line.

2. Write a code that prompts the user for a seed value and then generates a sequence of random numbers by rand() with this seed value until a duplicate of the last 5 digits is generated. Report this duplicated last 5 digits and its two occurrences. For example, for the following sequence of the last 5 digits of random numbers, your program should output

```
10113 repeats: 3  11
```

Because you don't know when the duplicate will be, you must use dynamic memory allocation to store all the distinct random numbers (or at least the last 5 digits). If you declare an array of huge size (for example RAND_MAX) to hold all the numbers generated by rand(), you will receive 0 (zero) point. Please generate exactly the output as the sample executable 4.2.out posted in public/ directory does. Detailed information of the random numbers can be obtained by running executable 4.2.detail.out with the same seed value.

```
 1:  16838
 2:   5758
 3:  10113
 4:  17515
 5:  31051
 6:   5627
 7:  23010
 8:   7419
 9:  16212
10:   4086
11:  10113
```